PCA

INFO 521 · Module 7 — Principal Component Analysis & Course Synthesis

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Adapted with permission from lecture materials by Clayton Morrison, University of Arizona.

Learning outcomes

By the end of Module 7 you should be able to:

1. **Compute PCA** — center, covariance, eigendecomposition — from scratch.
2. **State both views** — maximum variance $\iff$ minimum reconstruction error.
3. **Read variance explained** ($\lambda_k / \sum_j \lambda_j$) and choose a number of components with a scree plot.
4. **Interpret components on NHANES** — and state PCA's limits honestly.

The question: which directions matter?

Six NHANES features per patient — but are they six *independent* pieces of information?

- BMI and waist circumference track each other at $r \approx 0.90$ — nearly one measurement wearing two names.
- **PCA asks**: find new axes — **principal components** — that are *uncorrelated*, ordered by how much variance each carries.

$$\mathbf{z}_n = \mathbf{V}^\top \mathbf{x}_n \qquad \text{with } \mathbf{V} = [\mathbf{v}_1 \cdots \mathbf{v}_D] \text{ orthonormal.}$$

Keep the first $K \ll D$ coordinates: a **shorter description** of every patient, losing as little as possible.

Center first, then covariance

PCA is a statement about **spread around the mean**, so:

1. **Center** each feature ($\mathbf{x}_n \leftarrow \mathbf{x}_n - \bar{\mathbf{x}}$) — otherwise the first component just points at the mean.
2. **Standardize** when units differ (they do here — years, kg/m², cm, mg/dL, %): variance is unit-squared, and PCA maximizes it. m6b's scaling lesson, verbatim.
3. Form the **sample covariance** $\mathbf{C} = \frac{1}{N-1}\mathbf{X}^\top\mathbf{X}$ (centered $\mathbf{X}$) — the object m3b promised would return.

$$\mathbf{C}\,\mathbf{v}_k = \lambda_k\,\mathbf{v}_k \qquad \lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_D \geq 0$$

Eigenvectors = the new axes; **eigenvalues** = the variance each carries.

Two views, one solution

View 1 — max variance. Find the unit direction with the most spread:

$$\max_{\|\mathbf{v}\|=1} \ \mathbf{v}^\top \mathbf{C}\, \mathbf{v} \quad \Longrightarrow \quad \mathbf{v} = \mathbf{v}_1 \ \text{(top eigenvector)}.$$

View 2 — min reconstruction. Project onto K directions and rebuild; choose directions minimizing the squared error:

$$\min_{\mathbf{V}_K} \sum_{n=1}^{N} \left\| \mathbf{x}_n - \mathbf{V}_K \mathbf{V}_K^\top \mathbf{x}_n \right\|_2^2 \quad \Longrightarrow \quad \text{the same top-}K\text{ eigenvectors}.$$

Variance kept + reconstruction error = total variance (a constant): maximizing one **is** minimizing the other. Least squares (m1b’s projection!) closes the course as it opened it.

PC1 on the adiposity plane

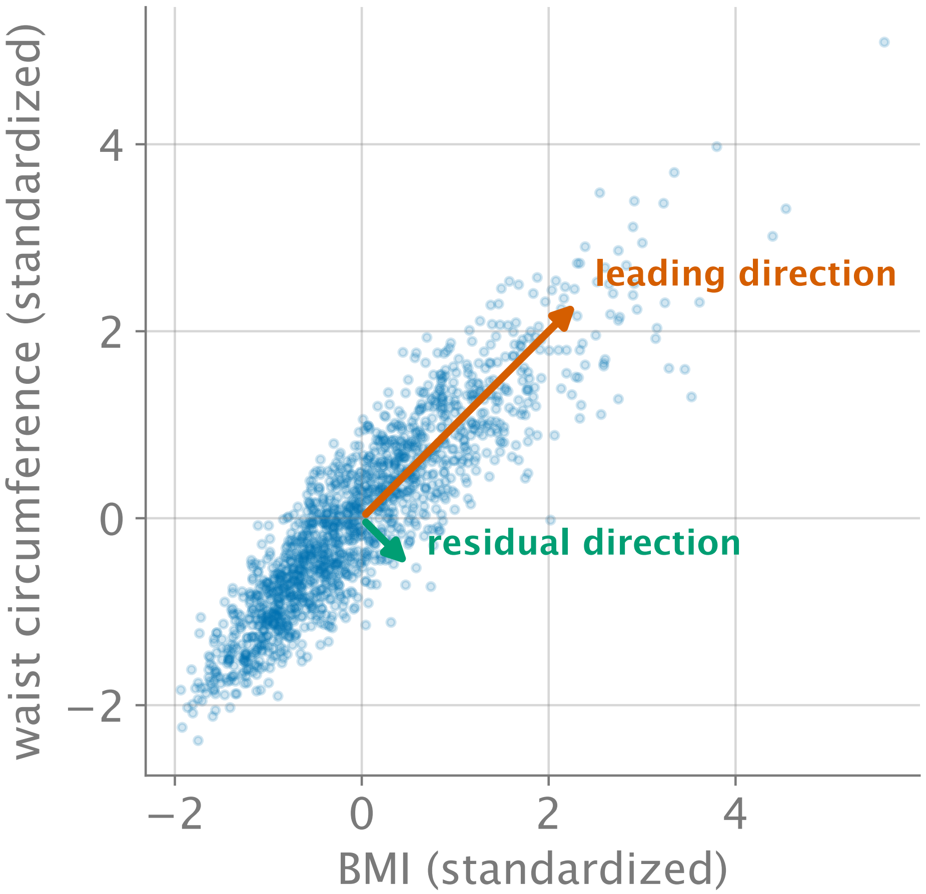


Figure 1: The (BMI, waist) plane, r ≈ 0.90: almost all spread lies along one shared direction (vermillion) — the redundancy PCA is built to remove.

All six features: the scree plot

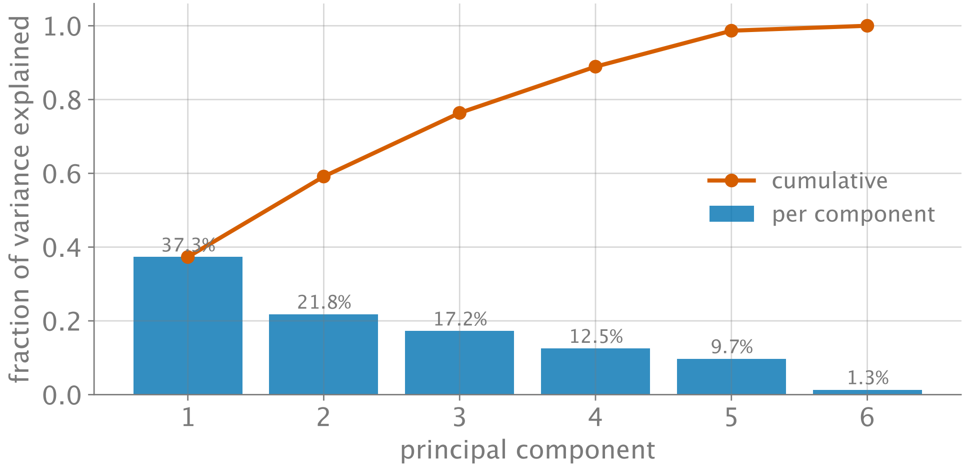


Figure 2: PCA on all six standardized features: PC1 carries 37.3%, PC1–2 59.1%, PC1–3 76.4% — and PC6 a residual 1.3% (the BMI/waist redundancy, found).

- **Variance explained**: $\lambda_k / \sum_j \lambda_j$.
- The near-zero PC6 (1.3%) is the algebra's way of saying *one of these six numbers is almost redundant* — we already knew which pair.

What do the components mean?

Loadings of the first two components (signs fixed by convention):

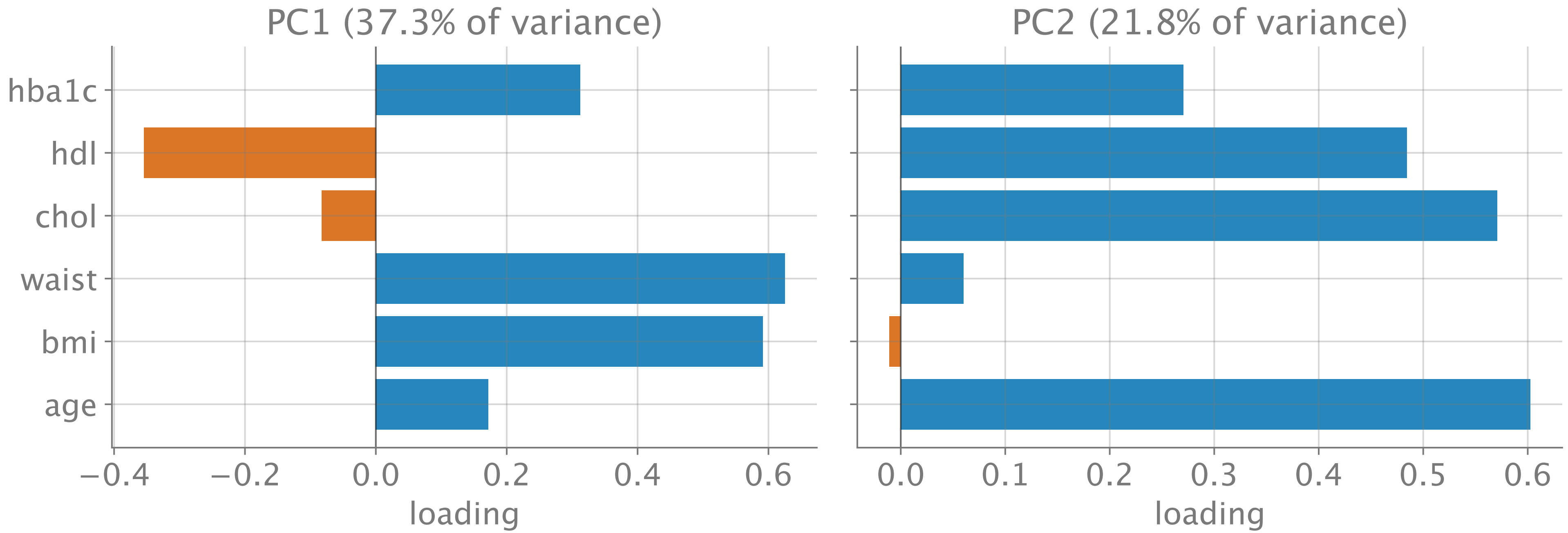


Figure 3: Loadings: PC1 ≈ an adiposity/metabolic axis (BMI + waist + HbA1c, HDL opposing); PC2 ≈ an age–cholesterol axis.

Choosing K, and what "explained" buys

- **Scree / elbow** — the m6b judgment call again: after PC3 the gains flatten; 2–3 components describe these patients reasonably.
- **Threshold** — "keep 90%" $\rightarrow$ here $K = 5$ (cumulative 98.7% at 5, 88.9% at 4).
- **Downstream use** — if the scores feed a supervised model, choose K by **cross-validation** on that model (m2a closes the loop).

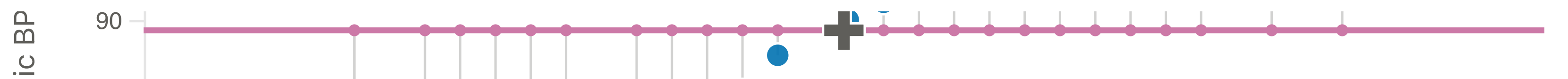
Compression is a *tool*, not a truth: variance explained is measured on $\mathbf{X}$ — no label ever voted.

Limitations — said plainly

- **Linear**: components are straight axes; a curved manifold (spirals, clusters on a ring) is invisible to them.
- **Variance ≠ usefulness**: PCA never saw the label. The direction that predicts hypertension may hide in a low-variance component — m6b's cluster-vs-label caution, transposed to axes.
- **Scale-dependent**: standardize or the units choose your components (m6b).
- **Interpretability is a gift, not a guarantee**: PC1 happened to be clinically readable here; rotated blends often aren't.

Live demo: the PCA explorer

Rotate the candidate axis in the week06 tool and watch variance-captured and reconstruction-error trade places — the two views of one solution.



Explorer — static fallback

The iframe above does not print or export to PDF; this snapshot stands in.

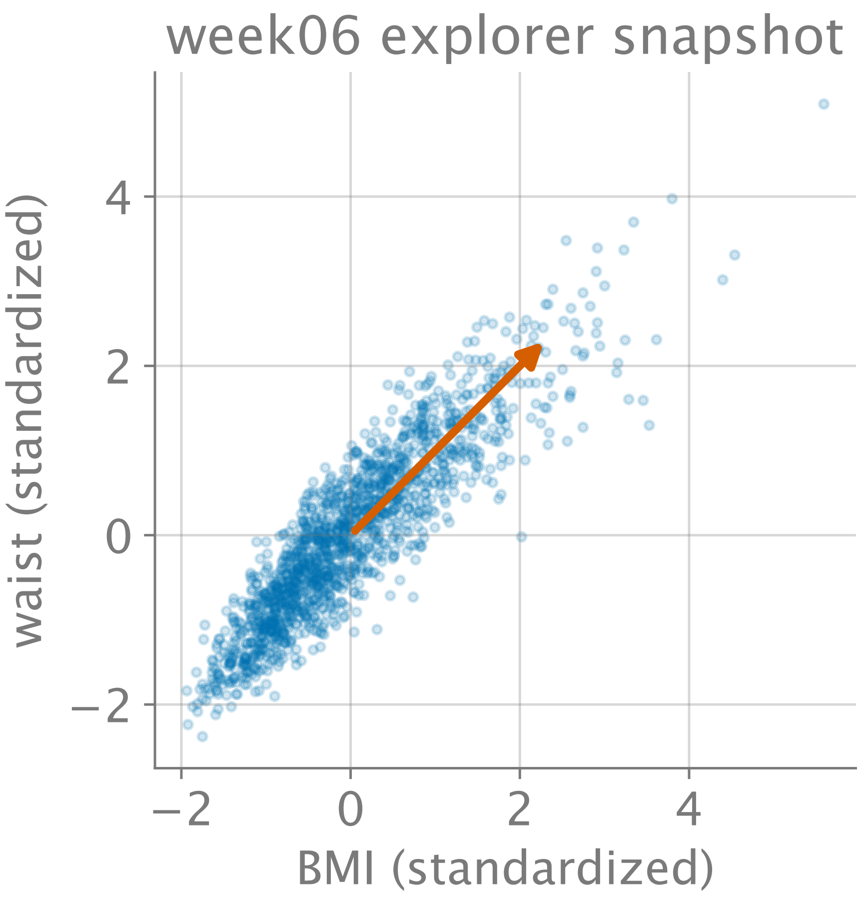


Figure 4: Static fallback: the leading direction on the adiposity plane (used in print / PDF).

The course, in one arc

One dataset, five ways of seeing — each module's answer became the next one's question:

	The move	The object
Loss (M1–M2)	minimize $\mathcal{L}$; control flexibility	$\hat{\mathbf{w}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$
Likelihood (M3)	model the noise; maximize ℓ	same $\hat{\mathbf{w}}$ + $\hat{\sigma}^2$ + $\mathrm{Cov}[\hat{\mathbf{w}}]$
Posterior (M4)	believe, then update	$\mathcal{N}(\mathbf{m}_N, \mathbf{S}_N)$ — Project 1's endpoint
Approximation (M5)	when conjugacy breaks, approximate	Laplace $\mathbf{A}^{-1}$; Metropolis draws
Discovery (M6–M7)	drop the labels	clusters; principal axes — described, not diagnosed

And one object followed you the whole way: $\mathbf{X}^\top \mathbf{X}$ — loss bowl, likelihood curvature, Fisher information, posterior precision, covariance. **That's the course.**